Machine Learning Project Report - Miraç Aydın

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The review video link: https://youtu.be/PCzGztIKD78

We have 2 datasets, one is called breast cancer dataset and the other one is wine dataset. Our purpose is to detect malignant cancer cells and cluster wine types. And we use Decision Tree, KNN and Naive Bayes algorithms for this.

Figures:

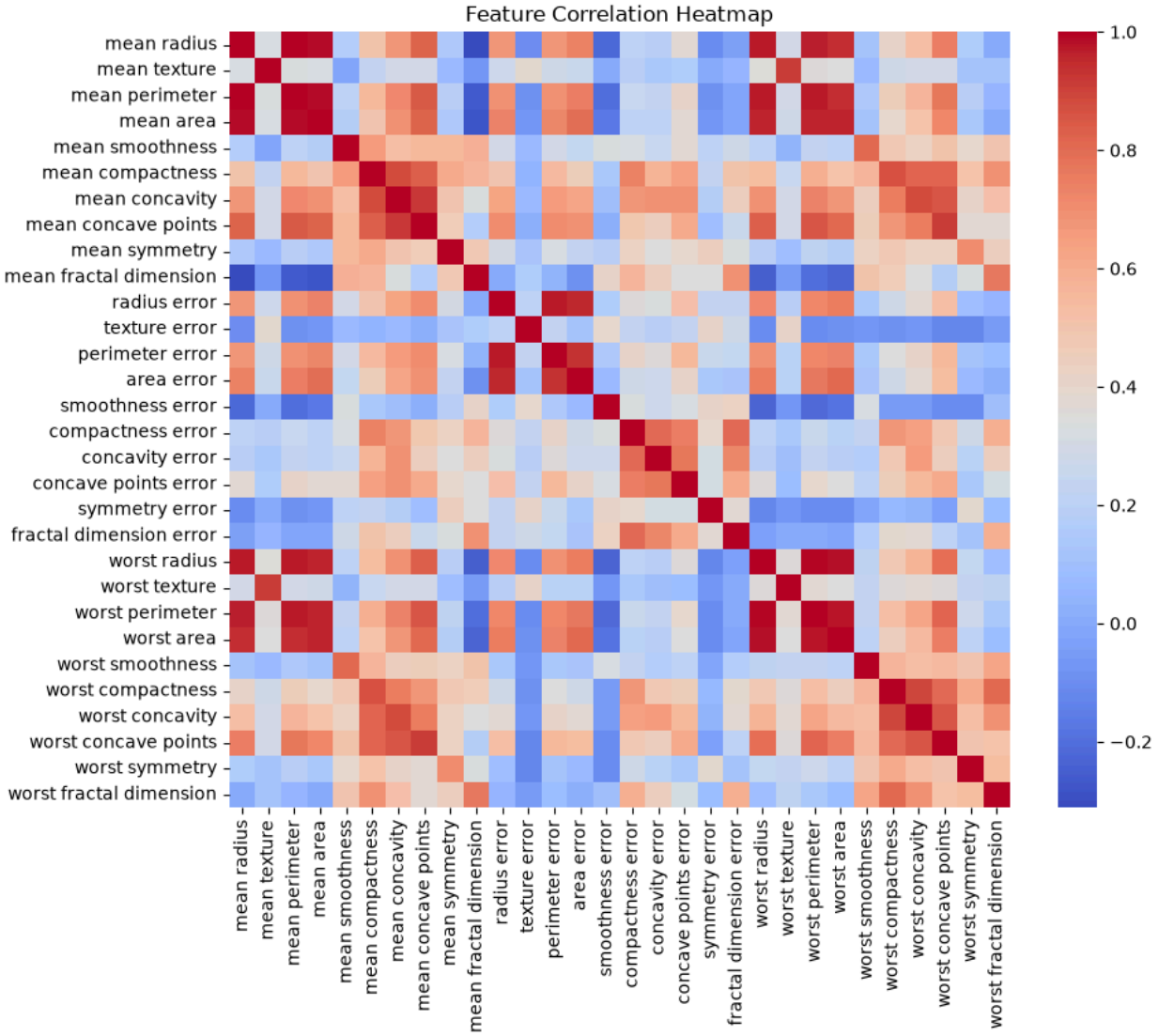
Our first figure is a CountPlot shown as below:

CountPlot:



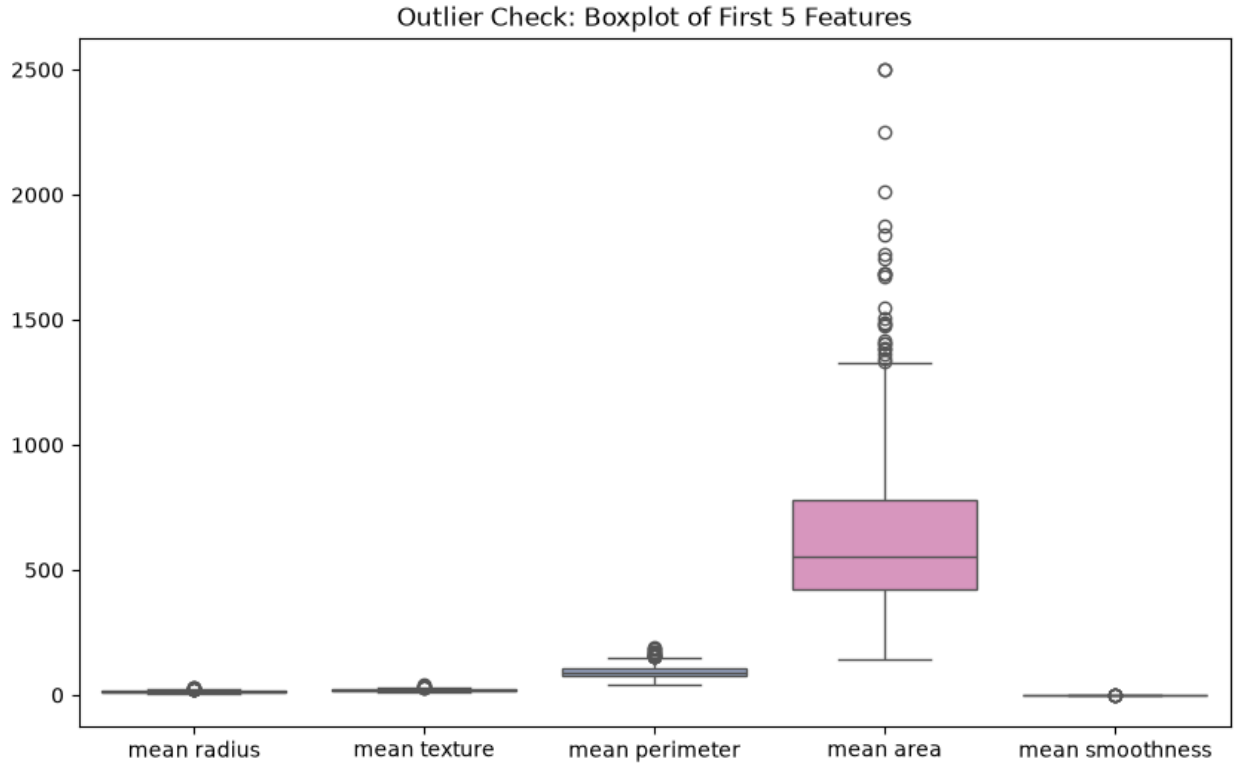
This plot checks if the dataset is balanced like how many malignant and bening exists.

The second figure is correlation heatmap.



This correlation heatmap shows how features are related to each other. Dark red and dark blue means high correlation.

The third one is Boxplot. We only used first 5 features, otherwise it would be too big to fit here.



This boxplot shows the distribution of the dataset(which must be numerical). As you can see there are many outliers on mean area and its body is much bigger than odders. Which means it needs scaling. So we used StandartScaler method.

To avoid data leakage the test set which consists of 20% of the total data is isolated and can't be read or accessed by our codes until the end. The scaler was fitted exclusively on training sets and only used to transform validation and test sets.

We use 3 different seeds to avoid the model being successful by total coincidence. The 3 different seeds are 42, 52 and 62 as specified.

Results:

22 ROW SUMMARY TABLE (Top 5)

	Algorithm	Representation	Parameter	Mean_Val_Macro_F1	Std_Val_Macro_F1	Mean_Val_Accuracy
16	knn	scaled	k7	0.967675	0.029718	0.970760
13	knn	scaled	k11	0.960846	0.039478	0.964912
14	knn	scaled	k3	0.954677	0.022757	0.959064
15	knn	scaled	k5	0.954677	0.022757	0.959064
12	knn	scaled	k1	0.936898	0.021295	0.941520

BEST MODEL:
Algorithm: knn, Scale: scaled, Setting: k7

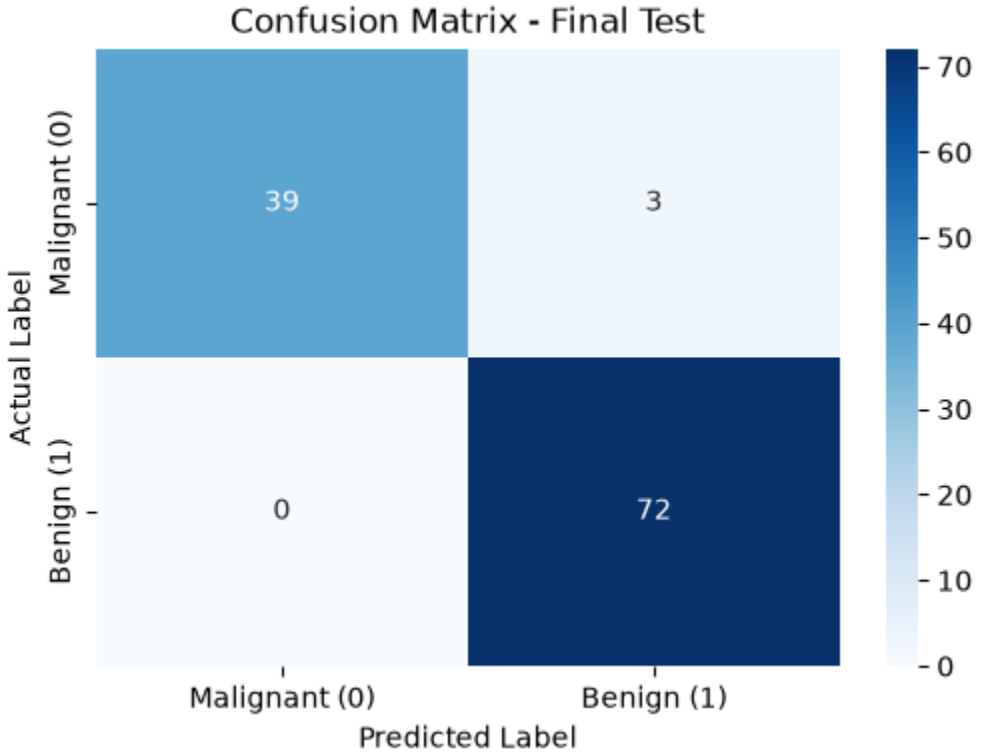
Our summary table shows best models for breast cancer data. Greater Mean_Val_Macro_F1, Mean_Val_Accuracy and Lesser Std_Val_Macro_F1, as stated in the code, can bee seen on the table.

22 ROW SUMMARY TABLE (Top 5)

	Algorithm	Representation	Parameter	Mean_Val_Macro_F1	Std_Val_Macro_F1	Mean_Val_Accuracy
13	knn	scaled	k11	0.981352	0.032299	0.981481
15	knn	scaled	k5	0.981352	0.032299	0.981481
16	knn	scaled	k7	0.981352	0.032299	0.981481
12	knn	scaled	k1	0.965812	0.029608	0.962963
10	gnb	scaled	default	0.963844	0.031359	0.962963

BEST MODEL:
Algorithm: knn, Scale: scaled, Setting: k11

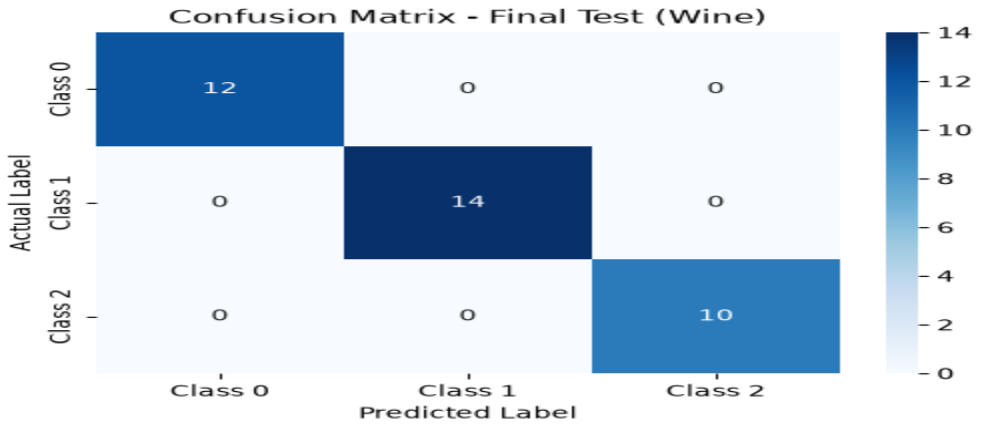
And this one shows the best models for wine data. Same stuff like breast cancer data(greater, greater, lesser).
As for the confusion matrix, on breast cancer data we have:



As you can see, our models priority is to decrease the number of False Negatives which is most important of all, you wouldn't want to miss a real cancer patient by assuming they are healthy. In this case we have 3 only.

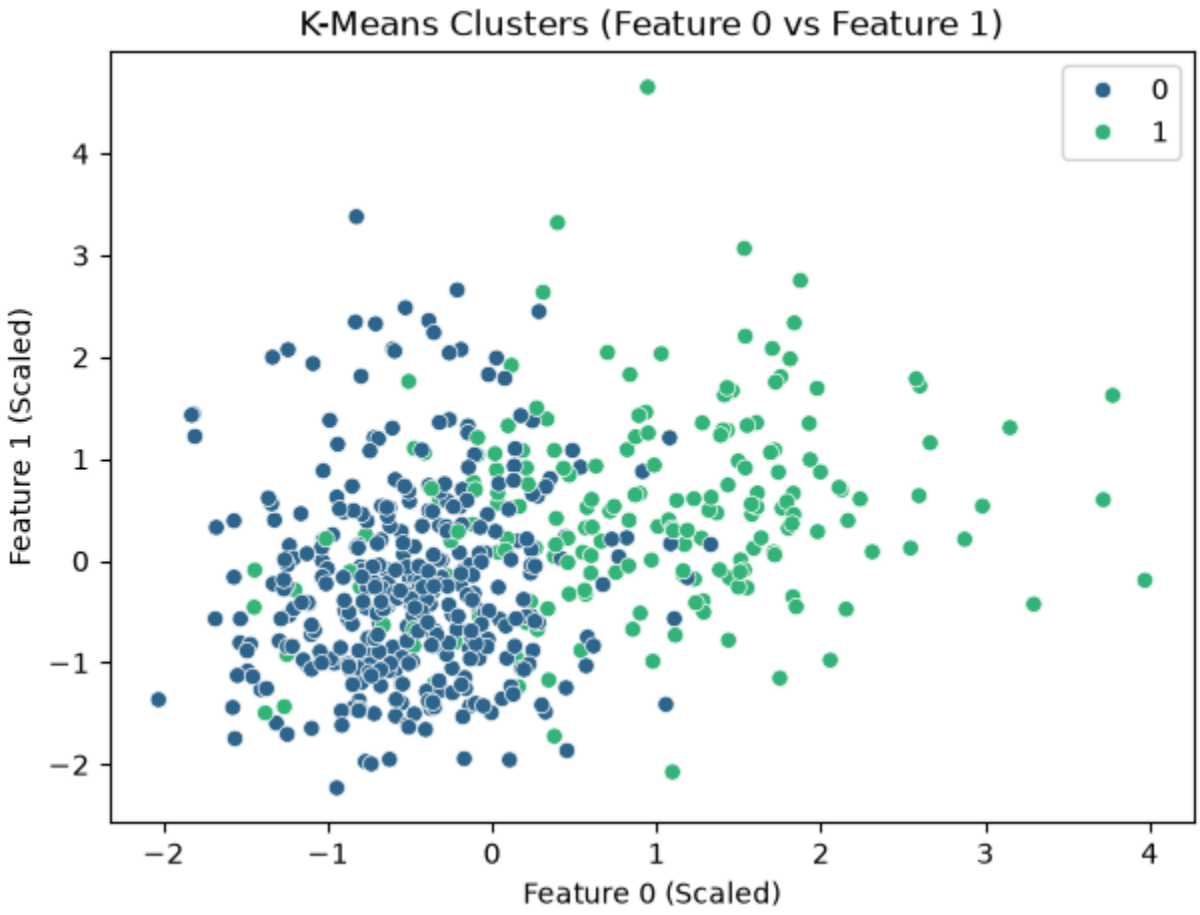
As for the wine data we have:

3 Different wine classes a



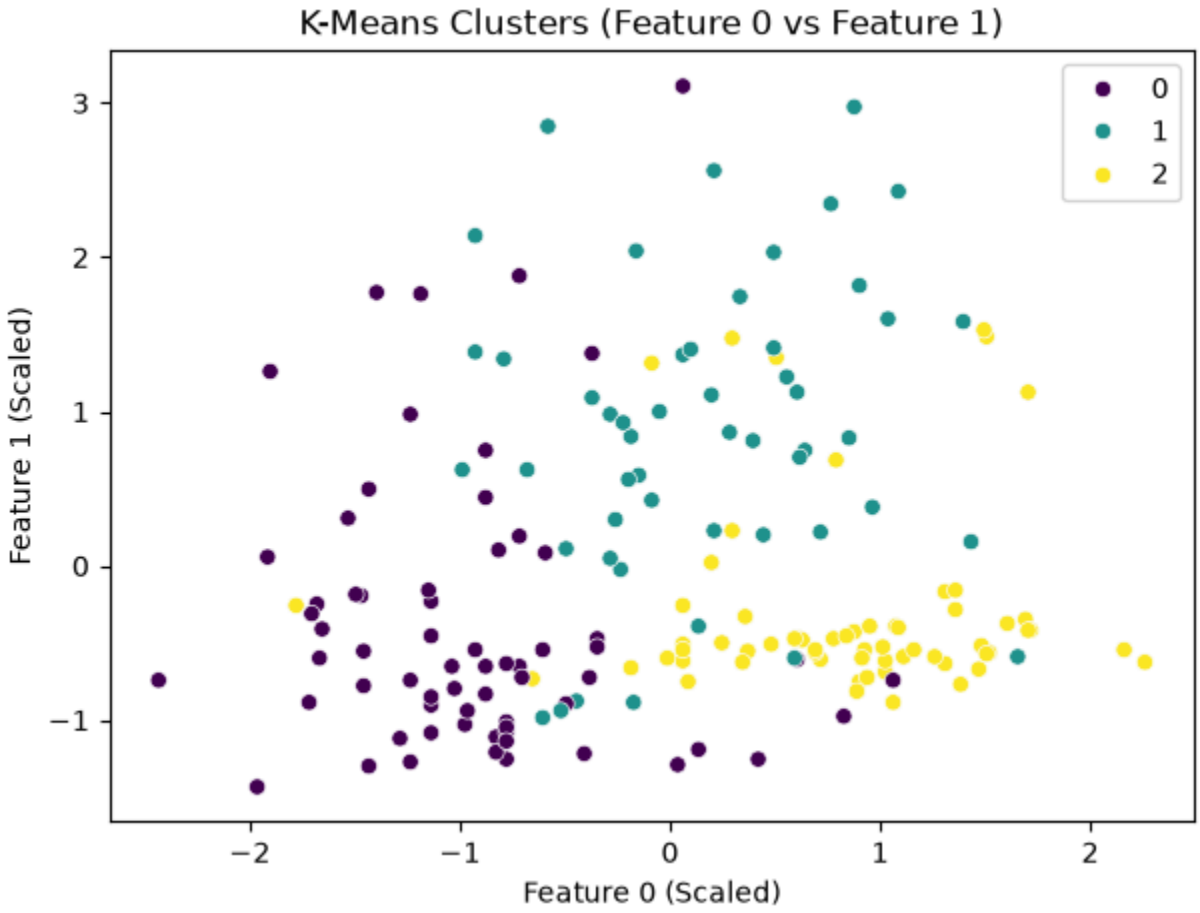
And its precision is pretty high.

Now K-Means Graphics:



This breast cancer k-means unsupervised learning results give us the following data:
Rand Index: 0.8279 — Silhouette Score: 0.3434

As for the wine:



Our results are:
Rand Index = 0.9543 — Silhouette Score: 0.2849

On the K-Means part, our model can't see the y(target) values, just by looking at the features of cell/wine, it got precision of %X on the Rand Index part as stated above on both graphs. You can see the precision of wine is higher(0.9543 > 0.8279). And this closes it